Vincent T. Cook

ROBOTICS & MECHATRONICS

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Aerospace engineering senior with robot circuitry design and repair, battery testing, CAD assemblies, and EV powertrain experience. Project work spans robot programming, 3D dynamics, and MATLAB/Simulink controls.

EDUCATION

San Jose State University | San Jose, CA
B.S. Aerospace Engineering | Expected May 2027 | Enrolled Sep 2022

TECHNICAL SKILLS

Design: SolidWorks, Autodesk Inventor, CAD, circuit troubleshooting, battery testing, stress analysis
Analysis: MATLAB/Simulink, MotionGenesis, Python, C++, Java, simulation, CFD, thermal testing

ENGINEERING EXPERIENCE & PROJECTS

SJSU Robotics | Mechanical Designer | Aug 2024-Present
- Designed a transmitter mast with camera/sensor mounts; used stress analysis to reinforce high-load regions and select lighter materials elsewhere.

FIRST Robotics Team 3925 | Electrical Lead - High School Robotics | 2020-2022
- Led robot circuitry design, battery testing, and circuit repairs; developed programming knowledge alongside electrical work.

SAE Spartan Racing | EV Powertrain Intern | Aug 2023-May 2025
- Created CAD and assembly instructions for an electric motor test bench; integrated mechanical/electrical components and coordinated engineering/manufacturing requirements.

Hybrid Electric / Hydrogen Fuel-Cell Aircraft | SJSU Capstone
Aerospace Engineering Team Member - In Progress | 2026-Present
- Contribute wing-model designs and project planning. Planned work includes CFD/thermal analysis, fabrication, and battery/fuel-system electronics; team flight target spring 2027.

BROADSWORD | Senior Design Rocket Concept | Mechanical & Programming Subgroup Lead | Aug-Dec 2023
- Designed landing legs in Inventor and coded stage liftoff/landing trajectories for a student rocket reusability concept; coordinated mechanical/programming tasks.

NASA L'SPACE Mission Concept Academy
Mechanical Engineering Team Member - Educational Program | Fall 2026-Present
- Help plan science approaches and evaluate mechanical components for a robotic lunar payload mission concept.

SELECTED COURSEWORK

- ENGR 10 / AE 20 | Designed, built, and programmed a task robot; modeled a V6 piston engine in Inventor and produced paper schematics.
- AE 138 / AE 140 / AE 30 | Used MotionGenesis for multibody orbital/Frisbee trajectories and MATLAB for Frisbee dynamics with external variables and moments of inertia; coded missile trajectories.
- AE 157 | Used MATLAB/Simulink for control-system projects.

WORK EXPERIENCE

PayPal Stadium | Guest Representative | 2026 Soccer Season
- Worked San Jose Earthquakes and Bay FC women's team matches.

The Tech Interactive | Educator I | 2024-2025
- Taught STEM labs/design challenges and explained technical concepts to diverse learners.

Self-Employed | Private Swim Instructor | 2023
- Scheduled clients and tailored swim lessons to individual skills and goals.

YMCA & Ventura Aquatics Center | Swim Instructor / Coach / Lifeguard | 2021-2024
- Coached swimmers, led practices, and adapted instruction for children with disabilities.

LEADERSHIP & ATHLETICS

SJSU Division I Club Ultimate | 2022-Present: coordinate gym training and conditioning. SJSU Club Swimming | Spring 2026: competed at a Cal-hosted meet.